



Functional Description

Project: “Advance Motors – Line 4 OP070”

Customer:

Advance Motors AB

Customer address:

Fabriksgatan 7, 54150

Skövde - Sweden

Customer reference:

AM0273



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1. Revision Control

1	Document created	M.A.	2024-12-06
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2. Functional Description

2.1. Overview

The proposed solution, shown in the figure below, includes 6 conveyors equipped with sensors running at 15m/min. Placed in the direction of production flow: one rotating conveyor will receive a given number of empty pallets and rotate them to face the rest of the conveyor line, this first conveyor will act as a pallet dispenser, forwarding a pallet at a time as they are needed. Following this conveyor are two conveyors that will serve as buffers, a fourth conveyor will keep the pallet being loaded with the rejected products stationary with a clamp, once full, the pallet will be forwarded through another conveyor serving as buffer, to a sixth conveyor, where it will wait for the operator-driven forklift to remove the pallet. The main HMI will be located near the security gate.

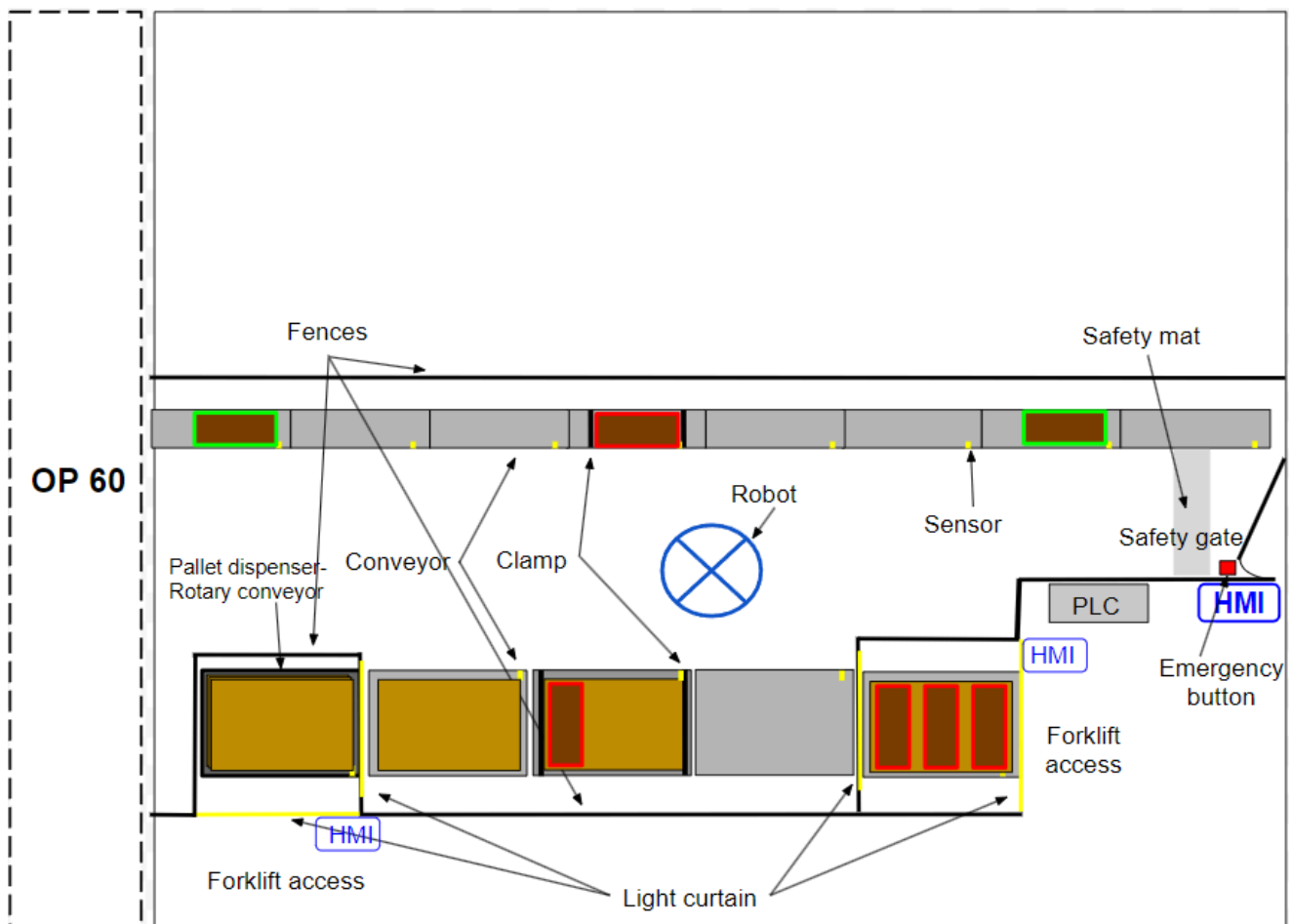


Figure 1: Design proposal



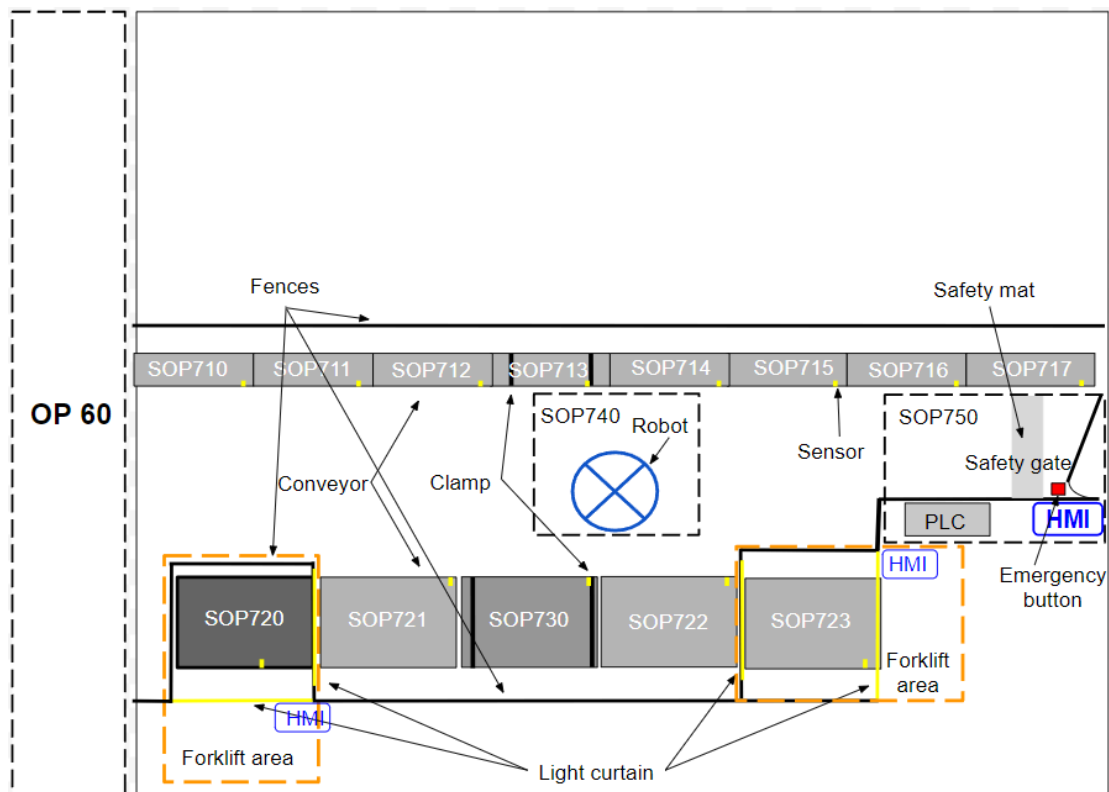
Fences (2.5m high) are located all around the cell, with the exception of the pallet supplier and removal areas. The access to the first and last conveyors are guarded by double action light curtains, they activate the outer curtains when no pallets are being introduced or removed, and activate the inner ones when the outer ones are disabled. This ensures that there is always a light curtain active between the inside and outside of the production line. As soon as an active light curtain is breached, the emergency stop is executed and the whole process stops. This ensures that no unauthorised elements enter the production area, possibly creating a hazardous situation. Beside both forklift access there will be light indicators and rope-switches to request access to the conveyors.

On the main conveyor line arriving from OP60, a clamp system is implemented in the conveyor from which the rejected pieces will be removed, this enables the system to stop the pieces in the exact same spot every time, ensuring repeatability.

2.2. Sub-Operations

The following picture shows how the operation is divided into different operations.

- SOP710-717: Conveyors 1 to 8 in the direction of the product flow where products arrive.
- SOP720-723: Conveyor 1, 2, 4 and 5 in the direction of the product flow through which pallets are transported. SOP720 and SOP723 include all devices in forklift areas.
- SOP730: Conveyor 3 in which defective details would be discarded.
- SOP740: Robot.
- SOP750: Safety area and gate. Includes the main cabinet, PLC and general devices.





2.3. Conveyor system

Each conveyor where products arrive will be commanded with a motor contactor to allow movement in both directions. Meanwhile, conveyors in which pallets are transported will be commanded with a single motor contactor to allow movement in only one direction. An inductive sensor will be used to detect the presence of details.

2.3.1. Production data

Production data will be following each detail through the line and will be transferred between each conveyor, even different operations. This production data includes 'Product Number', 'ID Number' and 'Status' information.

- Product Number (16 bits): It depends on the product model and is defined by the customer. The PLC uses just the last four digits, for example: MB16='1605', MB16T='1610', MB20='2004' and MB24='2407'.
- ID Number (32 bits): It is a unique number to identify each product (9 digits).
- Status (16 bits): Each bit in the status word is used to mark products status after different processes. OP060 will use bit 6 to mark the product as passed='1' or not passed='0' after inspection. Rest of the bits in the status word should not be changed, keeping their previous value before entering the operation.

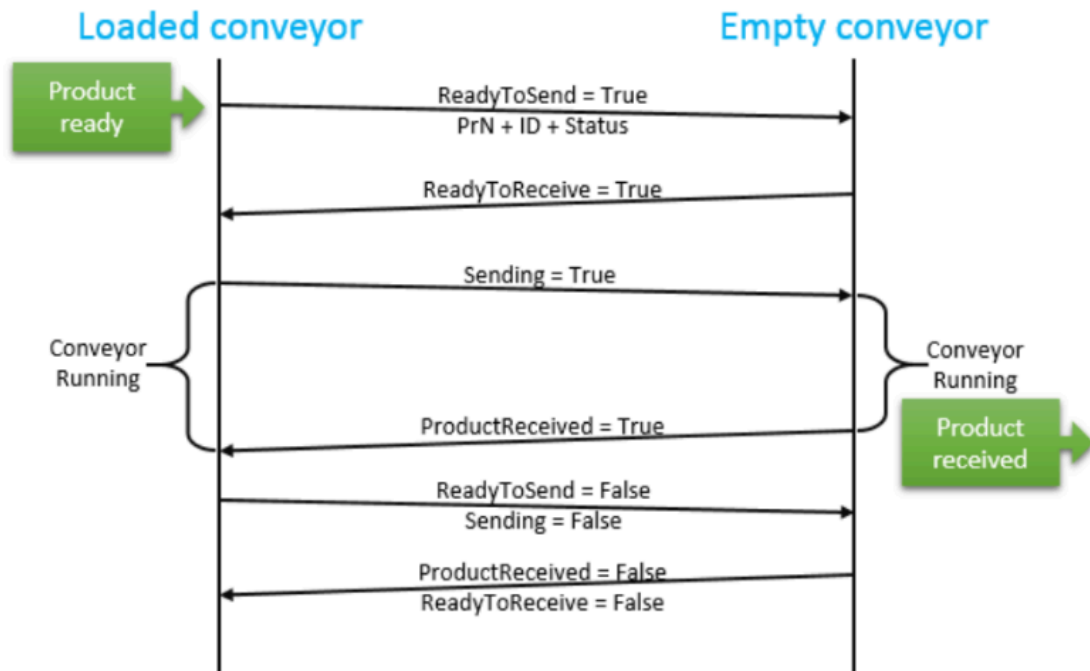
2.3.2. Product transferring

To ensure a safe and controlled product transfer between conveyors, a binary interlock protocol will be used. This protocol has been agreed with the customer and will be applied even between different operations.

Transferring between SOP710 to SOP717 will be programmed according to the following signals and flow.

Signals from loaded SOP	Data Type	Description
ReadyToSend	BOOL	SOP is ready to transfer a product
Sending	BOOL	SOP has started to transfer a product
PrN	UINT	Transferring product model (4 digits)
ID	UDINT	Transferring product ID (9 digits)
Status	WORD	Transferring product status 16-bits
Signals from empty SOP	Data Type	Description
ReadyToReceive	BOOL	SOP is ready to get a new product
ProductReceived	BOOL	SOP has received the product

NORMAL FLOW



2.3.3.SOP710 and SOP717: Input and output conveyor

SOP710 is the first conveyor in the OP70 and the interface with the previous operation OP60.

SOP717 is the last conveyor in the OP70 and the interface with the endline.

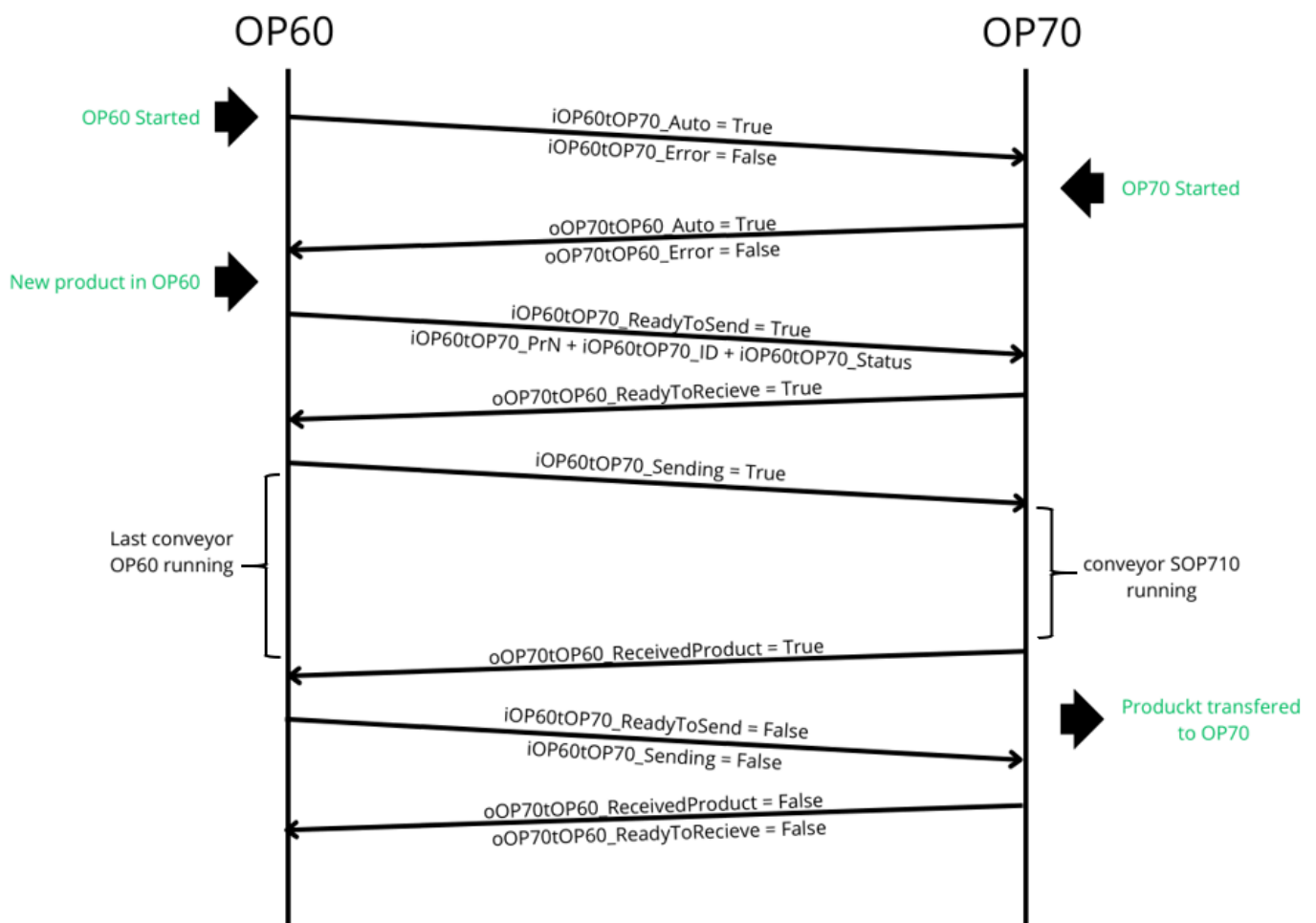
The signals and communication details are described below.

Inputs OP070	Data Type	Description
OP60tOP70_Auto	BOOL	OP60 is in Auto mode
OP60tOP70_Error	BOOL	OP60 has an active fault
OP60tOP70_ReadyToSend	BOOL	OP60 is ready to transfer a product
OP60tOP70_Sending	BOOL	OP60 has started to transfer a product
OP60tOP70_PrN	UINT	Transferring product model (4 digits)
OP60tOP70_ID	UDINT	Transferring product ID (9 digits)
OP60tOP70_Status	WORD	Transferring product status word
OP80tOP70_Auto	BOOL	OP80 is in Auto mode
OP80tOP70_Error	BOOL	OP80 has an active fault
OP80tOP70_ReadyToReceive	BOOL	OP80 is ready to get a new product
OP80tOP70_ReceivedProduct	BOOL	OP80 has received the product
Output OP60	Data Type	Description



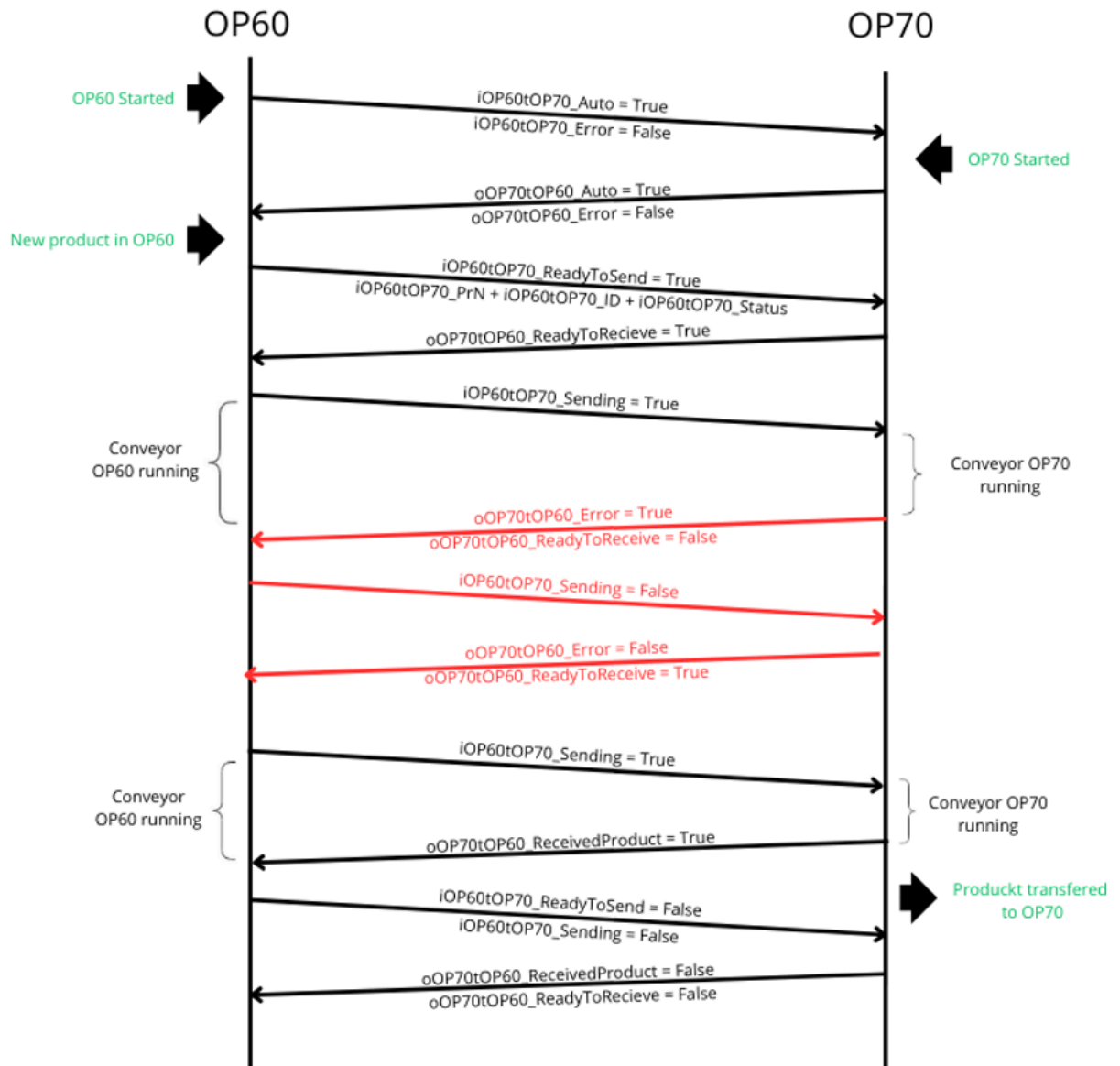
OP70tOP60_Auto	BOOL	OP70 is in Auto mode
OP70tOP60_Error	BOOL	OP70 has an active fault
OP70tOP60_ReadyToRecieve	BOOL	OP70 is ready to get a new product
OP70tOP60_RecievedProduct	BOOL	OP70 has received the product
OP70tOP80_Auto	BOOL	OP70 is in Auto mode
OP70tOP80_Error	BOOL	OP70 has an active fault
OP70tOP80_ReadyToSend	BOOL	OP70 is ready to transfer a product
OP70tOP80_Sending	BOOL	OP70 has started to transfer a product
OP70tOP80_PrN	UINT	Transferring product model (4 digits)
OP70tOP80_ID	UDINT	Transferring product ID (9 digits)
OP70tOP80_Status	INT	Transferring product status 16-bits

Shown below is the diagram of a normal transfer sequence:





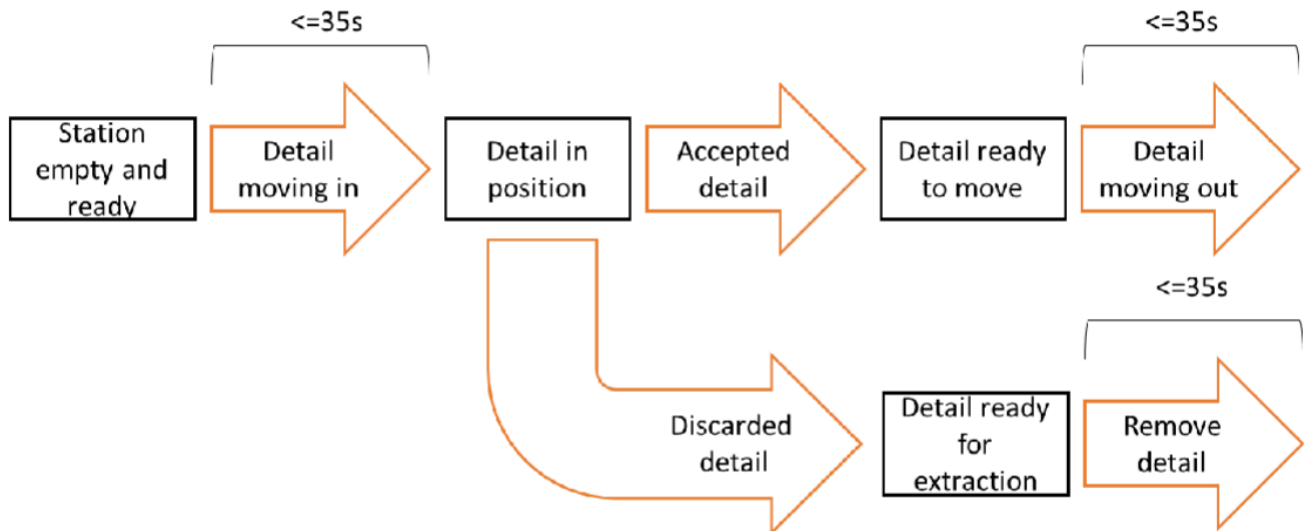
Below the diagram shows a faulty situation that stops the transfer. Note that both Auto and Error signals in both directions can break the transfer:





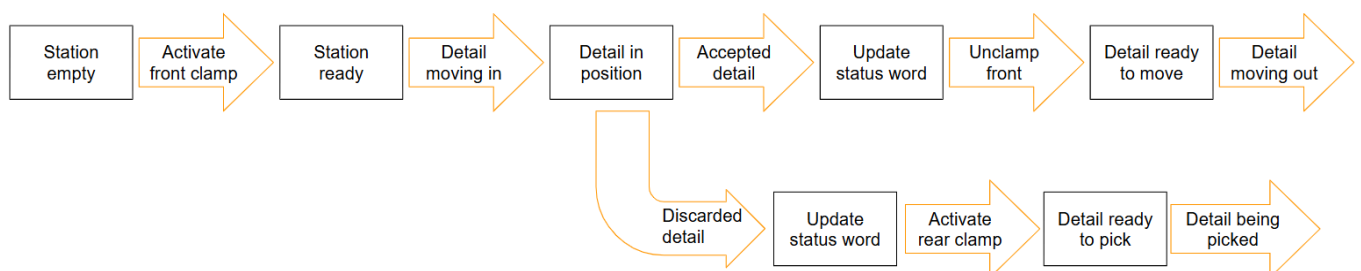
2.4. SOP713

SOP713 will be equipped with clamping devices in addition to the equipment included in the other conveyors. According to the customer specification the inspection station should follow the next working flow:



In order to fulfill the requirements, pneumatic clamping devices will be installed in SOP713, using double-acting cylinders. These devices will be acted with the corresponding valves and will clamp the details from the front and back. Sensors will be placed to detect both positions of each cylinder.

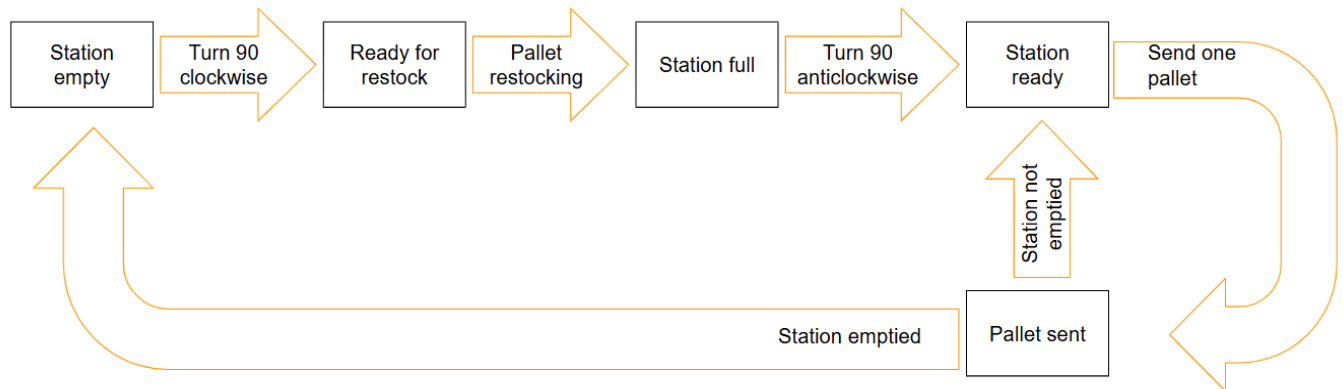
The detailed workflow is shown in the diagram below:





2.5. SOP720: Forklift area

SOP720 will consist of a rotatory pallet dispenser and light curtains in addition to the equipment included in the other conveyors.



When the station is empty, a signal will be shown in the HMI display to go and restock the pallets.

The pallet dispenser will be filled with an arbitrary amount of pallets, ie. 10, it will then rotate and face them all towards the conveyor flow. When a new pallet is needed, the dispenser will lift all the pallets except for the bottom one, which will be forwarded to the next buffer pallet.

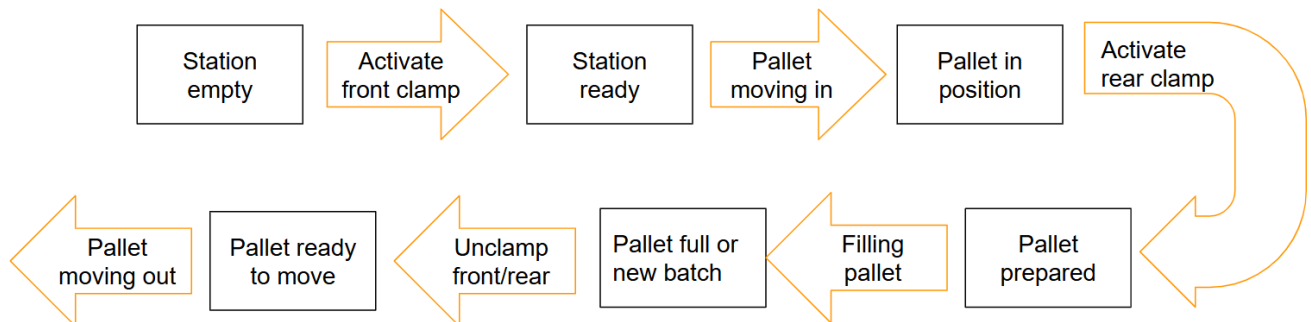
The interior light curtain will be muted during the pallet transfer from OP720 to OP721, during this transfer, the exterior light curtain will be activated. When the station is being restocked, the exterior light curtain will be muted and the interior one will be activated. The breach of an active light curtain will activate a safety fault, generating a safety stop.



2.6. SOP730

SOP730 will be equipped with clamping devices in addition to the equipment included in the other conveyors.

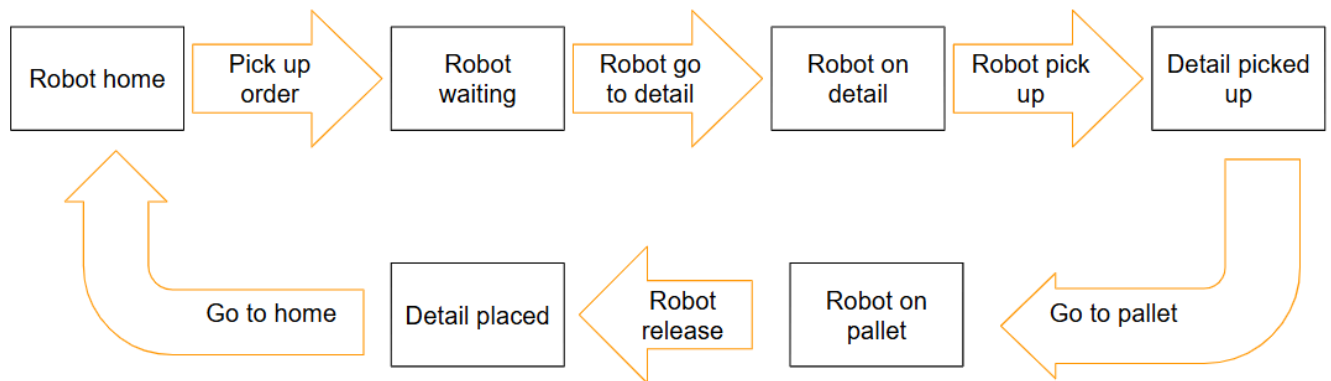
In order to fulfill the requirements, pneumatic clamping devices will be installed in SOP730, using double-acting cylinders. These devices will be acted with the corresponding valves and will clamp the pallets from the front and back. Sensors will be placed to detect both positions of each cylinder.





2.7. SOP 740

SOP740 corresponds to the operation of the robot. When the product is well placed and ready to pick up, the robot will receive a signal. When the signal is received, the robot should follow this workflow.



2.8. SOP750: Safety Gate and main cabinet

Safety gate and entrance to the safety area will be programmed according to the customer specifications. A safety mat will be located just inside the safety gate, it will be programmed in a way which will ensure that the process is always stopped if someone is inside the safety restricted area.

The main cabinet includes all the general devices, including control PLC, power supplies, main switches, protections, fuses, circuit breakers and motor contactors. It also includes all the pneumatic devices.

All protection and monitoring devices will be programmed according to the specifications to generate alarm indications if necessary (faults and warnings).